LQR Control of Downward Rotational Pendulum

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***Abstract*— The objective of this experiment was to implement and evaluate a full-state feedback controller on a downwards pendulum using both manual tuning and Linear Quadratic Regulator (LQR) methods. An LQR controller was created in MATLAB using the system's state-space model and iteratively refined Q and R matrices. Five test cases were performed to compare manual tuning, baseline LQR control, Q-matrix adjustments, and large decreases or increases in the R value. System performance was assessed using RMS platen and pendulum rotation errors, command effort, rise time and percent overshoot. The optimized LQR controller with adjusted Q values produced the best result, reducing pendulum RMS error from 21.65° to 12.96°, decreasing command effort by 20.9%, and lowering overshoot by 80.7%. In contrast, extreme modifications to the R value led to highly unstable or overly sluggish responses. These results demonstrate the sensitivity of LQR control to state and input weighting and highlight the importance of appropriately tuned LQR parameters for improved system behavior.**

***Index Terms*—Manual Tuning, LQR, Pendulum, Platen**

I. Introduction

THE main objective of this lab was to implement a full-state feedback controller on a downward rotational pendulum using manual tuning and LQR methods to have the platen trace a 60° square wave oscillating at 0.25 Hz while minimizing the pendulum rotation. A LabVIEW Virtual Instrument (VI) was set up with two proportional controllers and two derivative controllers corresponding to state space variables of the system. A series of manual tuning tests were conducted to determine control constant values that produced an acceptable response, and then an optimized LQR controller was implemented using MATLAB to iteratively test different values in the Q and R matrices until the best overall response was achieved.

The Furuta pendulum, consisting of a motor-powered rotating horizontal arm with a free-swinging vertical pendulum at its end, was selected for this experiment as it is a classic system used to analyze controller performance. Because the horizontal arm is the only directly controllable part of the system, it is a great platform for testing state space models and feedback control strategies. The Furuta pendulum can be in an upwards or downwards configuration and for this experiment only the latter was tested.

State space matrices for the Furuta pendulum system used in this lab were experimentally determined in Lab 3 as shown below [1].

$$[A] = [0,0,1,0;\ 0,0,0,1;\ 0,0.68,-18.3,0.02;\ 0,-36.8,8.11,-1.05] \tag{1}$$

$$[B] = [0,0,14.6,-6.46]^T \tag{2}$$

$$[C] = [1,0,0,0;\ 0,1,0,0] \tag{3}$$

$$[D] = [0] \tag{4}$$

To regulate the motion of the Furuta pendulum, a torque is applied to the horizontal arm based off an input sent to the motor defined as a weighted combination of the system states. In (5) below, k_1-k_4 are the adjustable gains of the system found either manually or using Bryson's rule then iteration. θ_1 is the angle of the arm, θ_2 is the pendulum angle, $\dot{\theta}_1$ is the velocity of the arm, and $\dot{\theta}_2$ is the velocity of the pendulum.

$$u = [k_1, k_2, k_3, k_4][\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2]^T \tag{5}$$

The main control method used in this lab was LQR which determines the optimal gain matrix by minimizing a cost function. The Q matrix shown in (6) is the state-weighting matrix which is diagonal to penalize deviations more strongly.

$$Q = \mathrm{diag}(Q_{11}, Q_{22}, Q_{33}, Q_{44}) \tag{6}$$

The Q matrix allows each state to be weighed independently, providing easy control over which should be prioritized. The R matrix in this experiment was a constant that inversely weighed the command effort input to the system.

II. Procedure

Materials and Experimental Setup

The required materials for this lab were a DAQ, laptop capable of running LabVIEW, a USB-C extension cord, tape, and the motor/flywheel system from Lab 1 with the attached downward pendulum that was modeled in Lab 3. The motor/flywheel and pendulum system consisted of a DC motor that rotated the platen in the horizontal plane with a gear set supported by ball bearings, and a rotational encoder that measured both the platen and pendulum angles to provide the real time state feedback needed to implement control strategies.

LabVIEW VI Setup

The LabVIEW VI framework from Lab 3 was modified to implement proportional and derivative control for both the platen and pendulum. Rotation errors were calculated by subtracting the desired values from the measured angles, with the pendulum reference set to 0°. These errors were multiplied

by gains k_1 and k_2 to provide proportional control, while the platen and pendulum angular velocities were multiplied by k_3 and k_4 to provide derivative control. The sum of these terms formed the motor command sent to the system. The counts per revolution were set to 2196 to match the gear ratio, the Turn On Command was set to 75, and the input signal was set to a 60° square wave at 0.25 Hz for all tests [2].

Manual Tuning Testing

A three-step manual tuning method was used to determine acceptable gains for the four controllers. First, the pendulum arm was taped in place so the platen could rotate without interference, and the pendulum gains k_2 and k_4 were set to zero. The platen gains k_1 and k_3 were increased gradually until oscillations occurred, then reduced until an acceptable platen response to the input wave was achieved. Next, the tape was removed, and the platen gains were set to zero so the pendulum rotation could be tuned independently. The gain k_2 was slowly increased and the pendulum was disturbed by hand to observe how effectively the system minimized its rotation, followed by adjustments to k_4 to improve damping. Finally, all four tuned gains were enabled together and refined further since their combined interaction initially caused instability and pendulum detachment. Gains were iteratively adjusted until both satisfactory input tracking and pendulum stabilization were achieved, at which point the data was saved for four cycles.

LQR Control Method Implementation

An LQR controller for the system was developed in MATLAB. First, the A and B matrices defining the system from Lab 3 were added to the code [1]. A 4x4 diagonal matrix with arbitrary starting values was defined as Q and a constant value was assigned to R. The four control constant values were then outputted by using the lqr() MATLAB function with the A, B, Q and R matrices as inputs. These control constant values were then input to the VI and tested. The Q and R values were then iteratively adjusted based off observations of what controls needed more emphasis until an acceptable response was achieved in terms of platen tracing and minimizing the pendulum rotation. Once the optimized LQR control constant values were found, data was saved for about four cycles.

Q Matrix Adjustment Test

The next task involved adjusting Q values of the previously attained LQR controller to achieve smaller pendulum angle deviations as the platen traced the input wave. To do this, the Q_{11} value was slightly decreased to reduce emphasis on platen rotation and the Q_{22} value was increased to place more emphasis on pendulum rotation. As a result, the output k_1 value decreased and the k_2 value increased, while the changes to k_3 and k_4 were minimal. The new control constant values were added to the VI and a test was initiated. Once smaller pendulum angle deviations were observed, four cycles of data were saved.

R Value Adjustment Tests

The final two tests involved greatly decreasing and increasing the R value while keeping the Q values the same as the previous test. First, the R value was set to 0.1 times its original value. A new set of control constant values were obtained from MATLAB and tested in LabVIEW. The motor command was so intense for this test that the pendulum had to be secured with tape in a way that still allowed it to rotate. Once secured, four cycles of data were taken. Next, the R value was set to 10 times its original value and the same process was repeated to save four more cycles of data. These tests showcased the impact of adjusting only the R value, or command effort, of the LQR controller.

Data Analysis

Data was saved for a total of 5 tests with the motor/flywheel and pendulum system—manually tuned method, LQR method, LQR with Q values adjusted, LQR with decreased R, and LQR with increased R. First, for every data set, a graph was made on Excel that plotted the platen rotation angle (°) and pendulum rotation angle (°) on the vertical axis against elapsed time in seconds on the horizontal axis. The two bests cycles were chosen, and all remaining data was deleted. Next, the root mean square, *RMS*, of the platen rotation error, pendulum rotation error and command values were then found using (3) below where N is the total data points and x_i is each individual error or command value [3].

$$RMS = \sqrt{(N^{-1})\Sigma(x_i)^2} \tag{3}$$

Finally, percent overshoot, $P.O.$, and rise time, t_r, were calculated for the platen rotation angle on Excel using the same methods outlined in Lab 2 [3]. Percent overshoot quantified how far the platen angle exceeded the desired rotation angle, while rise time quantified the speed of the response.

III. Results

One manual tuning test and four LQR method tests were conducted in this experiment. The LQR tests were based on system parameters characterized for the downward rotational pendulum in Lab 3. An impulse response test was conducted in MATLAB to ensure the parameters accurately represented the system as shown in Fig. 1 below.

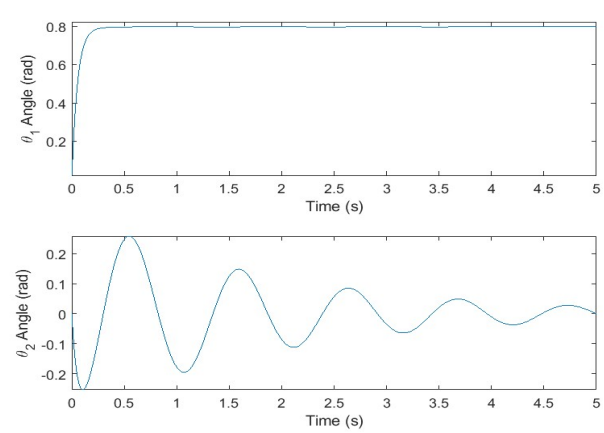

Fig. 1. Impulse response of platen angle (θ_1) and pendulum angle (θ_2) against time for pendulum down configuration plotted on MATLAB.

The response in Fig. 1 aligned with the expected behavior of a downward pendulum, verifying that the characterization in Lab 3 was done correctly. For every LQR test, a different combination of Q and R matrix values were used. Table I below shows the Q and R values for the four LQR tests where tests 2, 3, 4 and 5 correspond to the LQR, Q adjustment, decreased R and increased R test cases, respectively. Note that test 1 was the manual tuning method and was intentionally not included in the table below as Q and R matrices were not involved.

TABLE I
Q AND R MATRIX VALUES OF THE LQR TESTS

Test	Q_{11}	Q_{22}	Q_{33}	Q_{44}	R
2	100	200	0.1	0.1	6
3	80	250	0.1	0.1	6
4	80	250	0.1	0.1	0.6
5	80	250	0.1	0.1	60

Next, the resulting control constant values for the five test cases are included in Table II below where k_1, k_2, k_3 and k_4 are the control constants that correspond to platen rotation, pendulum rotation, platen velocity and pendulum velocity, respectively.

TABLE II
CONTROL CONSTANT VALUES OF ALL TEST CASES

Test	k_1	k_2	k_3	k_4
1	4.00	-3.00	0.20	0.15
2	4.08	-2.99	0.31	0.08
3	3.65	-3.61	0.30	0.07
4	11.6	-12.0	1.34	1.24
5	1.15	-0.58	0.07	-0.01

The control constant values from above were input to the VI for each test case to generate a response and record data for platen and pendulum rotation. The manual tuning response from test 1 is shown in Fig. 2 below.

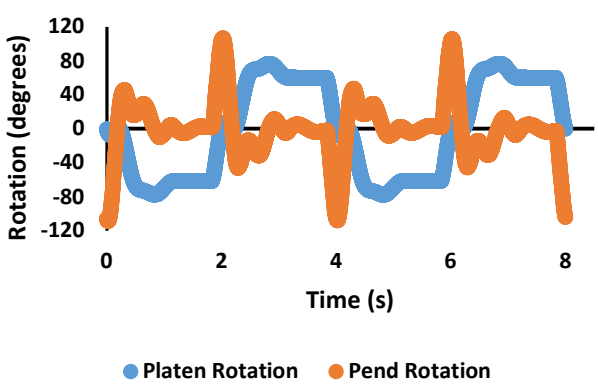

Fig. 2. Platen and pendulum rotation response for test case 1.

Next, the LQR method was used in test 2 to optimize the control constants and motor command. The response in Fig. 3 shows that the pendulum angle deviation decreased, exhibited by the smaller spikes of the orange data, but the platen tracing was not as accurate as test 1, exhibited by the increased overshoot of the blue data.

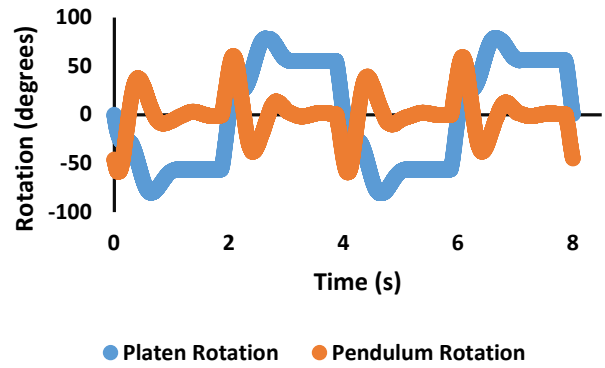

Fig. 3. Platen and pendulum rotation response for test case 2.

For test 3, the Q matrix values were adjusted to place more emphasis on decreasing pendulum angle variations. The response in Fig. 4 below reveals the method was successful as the pendulum angle deviations further decreased from test 2.

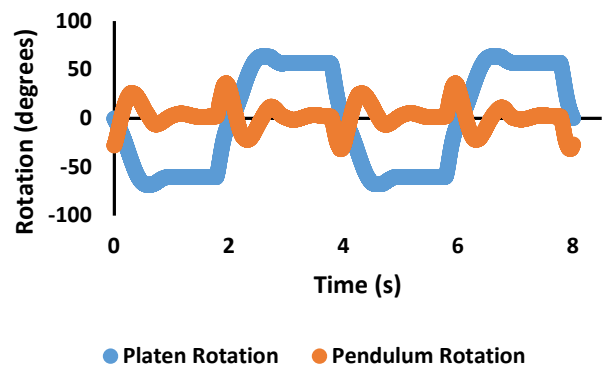

Fig. 4. Platen and pendulum rotation response for test case 3.

Test cases 4 and 5 featured multiplying the R value from test case 3 by a factor of 0.1 and 10, respectively, to show the effects of a greatly decreased and increased R value in LQR control. Fig. 5 below shows the erratic response from test case 4 where the motor command was too high to produce accurate results, and Fig. 6 shows the sluggish response from test case 5 where the motor command was too low for the platen to effectively trace the input wave.

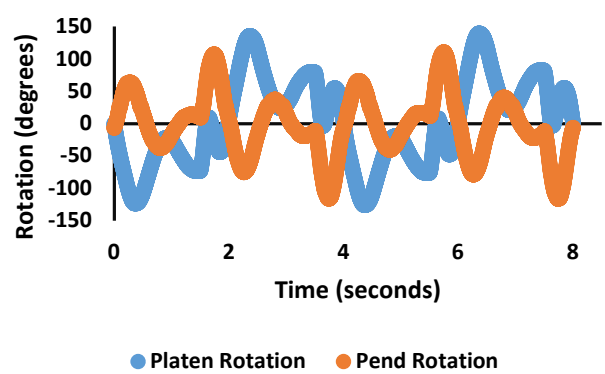

Fig. 5. Platen and pendulum rotation response for test case 4.

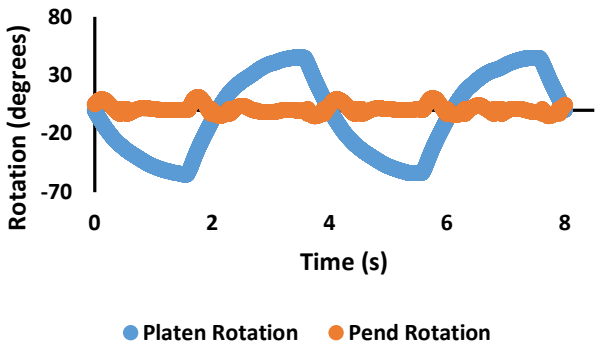

Fig. 6. Platen and pendulum rotation response for test case 5.

Based off the data presented above, the RMS of the platen rotation error ($RMS_{plat.err.}$), pendulum rotation error ($RMS_{pend.err.}$), and the command ($RMS_{comm.}$), as well as the rise time (t_r) and percent overshoot ($P.O.$) of the platen, were all calculated for each test case as shown in Table III below.

TABLE III
Q AND R MATRIX VALUES OF THE LQR TESTS

Test	$RMS_{plat.err.}$ (°)	$RMS_{pend.err.}$ (°)	$RMS_{comm.}$ (bits)	t_r (s)	$P.O.$ (%)
1	36.67	33.54	153.8	0.263	27.60
2	30.41	21.65	146.9	0.510	31.15
3	35.23	12.96	116.2	0.337	6.011
4	55.79	48.25	348.3	0.268	131.1
5	49.32	3.535	60.59	0.890	-24.86

IV. DISCUSSION

Restatement of Objectives

The objective of this lab was to implement a full-state feedback controller using both manual tuning and LQR methods to have the platen trace a 60° square wave oscillating at 0.25 Hz while minimizing pendulum rotation. Manual tuning was first used to identify reasonable control gains, after which an LQR controller was designed in MATLAB by adjusting Q and R matrices to achieve improved performance. Additional variations of Q and R were tested to observe how different weighting choices affected the system response.

Manual Tuning Method (Test 1) Versus LQR Method (Test 2)

The manual tuning and LQR method tests produced similar results due to the control constant values from test 1 being used as a guide when choosing arbitrary Q and R values, with the goal of achieving a more optimized combination of control constants. After analyzing the data from both test cases, the RMS values for the LQR test did reflect a more optimized response with a 16.47% decrease in platen rotation error, 35.45% decrease in pendulum rotation error and a 0.04% decrease in command effort. However, rise time and percent overshoot worsened for the LQR test with an increase in rise time by 93.92% and an increase in percent overshoot by 12.86%. The reason for the improved RMS values despite worse rise time and overshoot was presented visually in the plots for these test cases. For the LQR method to achieve significantly lower pendulum angle deviations, it had to overcompensate with more platen rotation before tracing the input wave leading to the increased rise time and overshoot.

Q Matrix Adjustment (Test 3) Analysis

The Q matrix adjustment test involved increasing Q_{22} (corresponding to pendulum rotation) by 25% and decreasing Q_{11} (corresponding to platen rotation) by 20% with the goal of further decreasing pendulum angle deviations from test 2. Not only was this achieved with a 40.14% decrease in RMS pendulum rotation error, but nearly every other performance metric improved as well with a 20.90% decrease in RMS command effort, a 33.92% decrease in rise time and an 80.70% decrease in percent overshoot. The only metric that got worse between tests 2 and 3 was RMS platen rotation error which increased by 15.85%, but this was expected with the decreased emphasis on platen rotation accuracy. Although the initial goal was to solely focus on improving pendulum angle deviations, test 3 featured the best response overall in terms of pendulum rotation, command effort and platen efficiency, showing the importance of placing high emphasis on this specific variable.

R Value Adjustments (Tests 4 and 5) Analysis

While tests 2 and 3 showed improvement from test 1, the opposite trend occurred in tests 4 and 5. In test 4, the R value from test 3 was multiplied by a factor of 0.1, which greatly increased command effort and caused a 199.74% rise in RMS command to 348.3 bits, along with an increase in RMS platen rotation error from 35.23° to 55.79°, a 272.30% increase in RMS pendulum rotation error, and a 2081.00% rise in percent overshoot, producing a response that was indistinguishable from the input wave. The only improvement was a 20.47% reduction in rise time due to the excessive motor effort.

In contrast, test 5 increased the R value by a factor of 10, significantly reducing command effort. The result was a sluggish response with a negative present overshoot and the slowest rise time of 0.89 seconds, which led to a 39.99% increase in RMS platen rotation error. However, the lower motor effort resulted in a 47.86% decrease in RMS command and a 72.72% reduction in RMS pendulum rotation error, as the platen moved too slowly to significantly disturb the pendulum. Overall, tests 4 and 5 produced the worst system performance, demonstrating that excessively low or high R values lead to unstable or overly sluggish behavior in LQR control.

Experimental Limitations and Suggestions for Improvement

The main limitation of this experiment was the hardware itself as the rough surfaces of the 3D printed bearings caused excess friction and noise, leading to a less responsive system overall. Replacing the 3D-printed bearings and mechanical interfaces with precision-machined components would reduce friction, minimize noise, and improve overall system responsiveness in future replications of this experiment.

REFERENCES

[1] Addeo Bryce, *Controls Lab Report 3.* Gainesville, FL, November 2025
[2] S. Ridgeway, *Lab #4 Assignment.* Gainesville, FL, November 2025
[3] Team 41, *Controls Lab Report 2.* Gainesville, FL, October 2025